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Studierichting: Informatica
Oefeningenreeks 4A nr. 30.2

$$\int x^3 \sqrt{x^2 + 1} dx$$

$$t = x^2 + 1$$

$$dt = 2x dx$$

$$\frac{dt}{2x} = dx$$

$$\begin{aligned} &= \int \frac{x^3 \sqrt{t} dt}{2x} \\ &= \int \frac{x^2 \sqrt{t} dt}{2} \\ &= \int \frac{(t-1) \sqrt{t} dt}{2} \\ &= \int \frac{(t-1) t^{\frac{1}{2}} dt}{2} \\ &= \frac{1}{2} \int \left(t^{\frac{3}{2}} - t^{\frac{1}{2}} \right) dt \\ &= \left(\frac{t^{\frac{5}{2}}}{\frac{5}{2}} - \frac{t^{\frac{3}{2}}}{\frac{3}{2}} \right) \frac{1}{2} + c \\ &= \left(\frac{2t^{\frac{5}{2}}}{5} - \frac{2t^{\frac{3}{2}}}{3} \right) \frac{1}{2} + c \\ &= \frac{t^{\frac{5}{2}}}{5} - \frac{t^{\frac{3}{2}}}{3} + c \\ &= \frac{\sqrt{t^5}}{5} - \frac{\sqrt{t^3}}{3} + c \\ &= \frac{\sqrt{(x^2+1)^5}}{5} - \frac{\sqrt{(x^2+1)^3}}{3} + c \end{aligned}$$

References